



TIA STANDARD

Project 25

ISSI Messages and Procedures for Voice Services, Mobility Management, and RFSS Capability Polling Services- Addendum 3- Interworking with an IWF

TIA-102.BACA-B-3
(Addendum to TIA-102.BACA)

April 2021

NOTICE

TIA Engineering Standards and Publications are designed to serve the public interest through eliminating misunderstandings between manufacturers and purchasers, facilitating interchangeability and improvement of products, and assisting the purchaser in selecting and obtaining with minimum delay the proper product for their particular need. The existence of such Standards and Publications shall not in any respect preclude any member or non-member of TIA from manufacturing or selling products not conforming to such Standards and Publications. Neither shall the existence of such Standards and Publications preclude their voluntary use by Non-TIA members, either domestically or internationally.

Standards and Publications are adopted by TIA in accordance with the American National Standards Institute (ANSI) patent policy. By such action, TIA does not assume any liability to any patent owner, nor does it assume any obligation whatever to parties adopting the Standard or Publication.

This Standard does not purport to address all safety problems associated with its use or all applicable regulatory requirements. It is the responsibility of the user of this Standard to establish appropriate safety and health practices and to determine the applicability of regulatory limitations before its use.

Any use of trademarks in this document are for information purposes and do not constitute an endorsement by TIA or this committee of the products or services of the company.

(From Standards Proposal No. TIA-PN-102.BACA-B-3, formulated under the cognizance of the TIA TR-8 Mobile and Personal Private Radio Standards, TR-8.19 Subcommittee on Wireline System Interfaces).

Published by
©TELECOMMUNICATIONS INDUSTRY ASSOCIATION
Technology and Standards Department
1310 N. Courthouse Road, Suite 890
Arlington, VA 22201 U.S.A.

**PRICE: Please refer to current Catalog of
TIA TELECOMMUNICATIONS INDUSTRY ASSOCIATION STANDARDS
AND ENGINEERING PUBLICATIONS
or call IHS, USA and Canada
(1-800-854-7179) International (303-397-2896)
or search online at www.TIAonline.org**

All rights reserved

Printed in U.S.A.

NOTICE OF COPYRIGHT

This document is copyrighted by the TIA.

Reproduction of these documents either in hard copy or soft copy (including posting on the web) is prohibited without copyright permission. For copyright permission to reproduce portions of this document, please contact the TIA Standards Department or go to the TIA website (www.tiaonline.org) for details on how to request permission. Details are located at:

[Standards Procedures and Guidelines](#)

or

Telecommunications Industry Association
Technology & Standards Department
1310 N. Courthouse Road, Suite 890
Arlington, VA 22201 USA
+1.703.907.7700

Organizations may obtain permission to reproduce a limited number of copies by entering into a license agreement. For information, contact:

IHS
15 Inverness Way East
Englewood, CO 80112-5704
or call
USA and Canada (1.800.525.7052)
International (303.790.0600)

Limited Use Copy

NOTICE OF DISCLAIMER AND LIMITATION OF LIABILITY

The document to which this Notice is affixed (the "Document") has been prepared by one or more Engineering Committees or Formulating Groups of the Telecommunications Industry Association ("TIA"). TIA is not the author of the Document contents, but publishes and claims copyright to the Document pursuant to licenses and permission granted by the authors of the contents.

TIA Engineering Committees and Formulating Groups are expected to conduct their affairs in accordance with the TIA Procedures for American National Standards and TIA Engineering Committee Operating Procedures, the current and predecessor versions of which are available at [Standards Procedures and Guidelines](#). TIA's function is to administer the process, but not the content, of document preparation in accordance with the Manual and, when appropriate, the policies and procedures of the American National Standards Institute ("ANSI"). TIA does not evaluate, test, verify or investigate the information, accuracy, soundness, or credibility of the contents of the Document. In publishing the Document, TIA disclaims any undertaking to perform any duty owed to or for anyone.

If the Document is identified or marked as a project number (PN) document, or as a standards proposal (SP) document, persons or parties reading or in any way interested in the Document are cautioned that: (a) the Document is a proposal; (b) there is no assurance that the Document will be approved by any Committee of TIA or any other body in its present or any other form; (c) the Document may be amended, modified or changed in the standards development or any editing process.

The use or practice of contents of this Document may involve the use of intellectual property rights ("IPR"), including pending or issued patents, or copyrights, owned by one or more parties. TIA makes no search or investigation for IPR. When IPR consisting of patents and published pending patent applications are claimed and called to TIA's attention, a statement from the holder thereof is requested, all in accordance with the Manual. TIA takes no position with reference to, and disclaims any obligation to investigate or inquire into, the scope or validity of any claims of IPR. TIA will neither be a party to discussions of any licensing terms or conditions, which are instead left to the parties involved, nor will TIA opine or judge whether proposed licensing terms or conditions are reasonable or non-discriminatory. TIA does not warrant or represent that procedures or practices suggested or provided in the Manual have been complied with as respects the Document or its contents.

If the Document contains one or more Normative References to a document published by another organization ("other SSO") engaged in the formulation, development or publication of standards (whether designated as a standard, specification, recommendation or otherwise), whether such reference consists of mandatory, alternate or optional elements (as defined in the TIA Procedures for American National Standards) then (i) TIA disclaims any duty or obligation to search or investigate the records of any other SSO for IPR or letters of assurance relating to any such Normative Reference; (ii) TIA's policy of encouragement of voluntary disclosure (see TIA Procedures for American National Standards Annex C.1.2.3) of Essential Patent(s) and published pending patent applications shall apply; and (iii) Information as to claims of IPR in the records or publications of the other SSO shall not constitute identification to TIA of a claim of Essential Patent(s) or published pending patent applications.

TIA does not enforce or monitor compliance with the contents of the Document. TIA does not certify, inspect, test or otherwise investigate products, designs or services or any claims of compliance with the contents of the Document.

ALL WARRANTIES, EXPRESS OR IMPLIED, ARE DISCLAIMED, INCLUDING WITHOUT LIMITATION, ANY AND ALL WARRANTIES CONCERNING THE ACCURACY OF THE CONTENTS, ITS FITNESS OR APPROPRIATENESS FOR A PARTICULAR PURPOSE OR USE, ITS MERCHANTABILITY AND ITS NONINFRINGEMENT OF ANY THIRD PARTY'S INTELLECTUAL PROPERTY RIGHTS. TIA EXPRESSLY DISCLAIMS ANY AND ALL RESPONSIBILITIES FOR THE ACCURACY OF THE CONTENTS AND MAKES NO REPRESENTATIONS OR WARRANTIES REGARDING THE CONTENT'S COMPLIANCE WITH ANY APPLICABLE STATUTE, RULE OR REGULATION, OR THE SAFETY OR HEALTH EFFECTS OF THE CONTENTS OR ANY PRODUCT OR SERVICE REFERRED TO IN THE DOCUMENT OR PRODUCED OR RENDERED TO COMPLY WITH THE CONTENTS. TIA SHALL NOT BE LIABLE FOR ANY AND ALL DAMAGES, DIRECT OR INDIRECT, ARISING FROM OR RELATING TO ANY USE OF THE CONTENTS CONTAINED HEREIN, INCLUDING WITHOUT LIMITATION ANY AND ALL INDIRECT, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES (INCLUDING DAMAGES FOR LOSS OF BUSINESS, LOSS OF PROFITS, LITIGATION, OR THE LIKE), WHETHER BASED UPON BREACH OF CONTRACT, BREACH OF WARRANTY, TORT (INCLUDING NEGLIGENCE), PRODUCT LIABILITY OR OTHERWISE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THE FOREGOING NEGATION OF DAMAGES IS A FUNDAMENTAL ELEMENT OF THE USE OF THE CONTENTS HEREOF, AND THESE CONTENTS WOULD NOT BE PUBLISHED BY TIA WITHOUT SUCH LIMITATIONS.

Limited Use Copy

Foreword

(This foreword is not part of this standard.)

This document has been submitted to APCO/NASTD/FED by the Telecommunications Industry Association (TIA), as provided for in a Memorandum of Understanding (MOU) dated December, 1993. That MOU provides that APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall provide technical assistance in the development of documentation for the Standard. This document has been developed by TR8.19 (Wireline Interfaces) with inputs from the APCO Project 25 Interface Committee (APIC), the APIC ISSI Task Group, and TIA Industry members. This document is being published to provide technical information on the emerging digital techniques for Land Mobile Radio Service.

The symbol "P25" or the wording "Project 25" appearing on the cover sheet of this document when published or in the title of any TIA-102 document or Telecommunications Service Bulletin (TSB) referenced herein, that indicates the APCO/NASTD/FED Project 25 Steering Committee has adopted the document as part of the Standard. The appearance of the abbreviation "P25" or the wording "Project 25" on the cover sheet of this document or in the title of any document referenced herein does not limit the applicability of the information contained in this document to "P25" or "Project 25" implementations exclusively.

This document is an addendum to TIA-102.BACA-B, November 2012, Project 25 Inter-RF Subsystem Interface (ISSI) Messages and Procedures for Voice and Mobility Management Services. The ISSI is the interface between Radio Frequency Subsystems (RFSSs) enabling relevant Project 25 user requirements to be satisfied (e.g., as described in the Inter-RF Subsystem Interface Overview (TSB-102.BACC-B)). This addendum defines TIA-102 and 3GPP MCPTT interconnectivity via an Interworking Function (IWF) for specific voice and mobility management services. The interconnection between the TIA-102 system and IWF is achieved via the ISSI. See the Annex for further details.

Patent Identification

The reader's attention is called to the possibility that compliance with this document may require the use of one or more inventions covered by patent rights. By publication of this document no position is taken with respect to the validity of those claims or any patent rights in connection therewith. The patent holders so far identified have, we believe, filed statements of willingness to grant licenses under those rights on reasonable and nondiscriminatory terms and conditions to applicants desiring to obtain such licenses.

The following patent holders and patents have been identified in accordance with the TIA intellectual property rights policy:

No patents have been identified.

TIA shall not be responsible for identifying patents for which licenses may be required by this document or for conducting inquiries into the legal validity or scope of those patents that are brought to its attention.

Trademark Identification

The following trademarks are referenced in this specification:

IMBE™ is a trademark of Digital Voice Systems, Inc.¹

¹ Use of the trademark identified in this standard is not an endorsement by TIA.

TABLE OF CONTENTS

A.	ADDENDUM INTRODUCTION.....	1
B.	ADDENDUM SCOPE	1
C.	ADDENDUM REFERENCES.....	1
C.1	NORMATIVE REFERENCES.....	1
C.2	INFORMATIVE REFERENCES.....	2
1.	INTRODUCTION	3
1.1	SCOPE	3
1.2	TERMINOLOGY	3
1.3	DOCUMENT STRUCTURE	3
1.4	DEFINITIONS.....	3
1.5	ABBREVIATIONS	3
1.6	REFERENCES	3
1.6.1	<i>Normative References</i>	<i>3</i>
1.6.2	<i>Informative References.....</i>	<i>3</i>
2.	OVERVIEW OF ARCHITECTURE AND PROTOCOL SUITE.....	5
3.	SIP MESSAGES AND PARAMETERS DEFINITION FOR ISSI	6
4.	RTP MESSAGE VOCABULARY FOR ISSI	7
5.	MOBILITY MANAGEMENT	8
6.	CALL CONTROL.....	9
7.	PUSH-TO-TALK (PTT) MANAGEMENT	10
8.	RFSS SERVICE CAPABILITY POLLING	11
	ANNEX A (NORMATIVE) – GENERIC SIP RESPONSES OVER THE ISSI.....	12
	ANNEX B (NORMATIVE) – TABLE OF TIMERS AND CONSTANTS	12
	ANNEX C (INFORMATIVE) – TIA-102 VOCODER ERROR MITIGATION DATA.....	13
	ANNEX D (INFORMATIVE) – SU-TO-SU CALL FLOWS.....	14
	ANNEX E (INFORMATIVE) – GROUP CALL FLOWS.....	15
	ANNEX F (INFORMATIVE) – SU-TO-SU CALL FLOWS INVOLVING MULTIPLE VOCODER MODES.....	16
	ANNEX G (INFORMATIVE) – GROUP CALL FLOWS INVOLVING MULTIPLE VOCODER MODES.....	17
	ANNEX H (INFORMATIVE) – FUTURE CONSIDERATIONS	18
	ANNEX I (INFORMATIVE) – CONSOLE FUNCTIONALITY	19
	ANNEX J (NORMATIVE) – INTERWORKING WITH AN IWF.....	20
J.1	INTRODUCTION	20
J.2	PURPOSE AND SCOPE OF THIS ANNEX	20
J.3	ARCHITECTURE	20
J.4	OVERVIEW	21
J.4.1	<i>System Diagrams with Consoles and CSSI.....</i>	<i>22</i>
2.	OVERVIEW OF ARCHITECTURE AND PROTOCOL SUITE.....	25

2.1	ARCHITECTURE OVERVIEW	25
2.1.1	<i>Call Models</i>	26
2.1.1.1	Losing Audio Scenarios.....	26
2.1.1.2	Confirmed and Unconfirmed Group Calls	26
2.1.2	<i>Subsystem Topologies</i>	27
2.1.3	<i>Functional Model</i>	28
2.1.3.1	Mobility Management.....	29
2.1.3.2	Call Control.....	31
2.1.3.2.1	Unit Call Control.....	31
2.1.3.2.2	Group Call Control	32
2.1.3.3	Media Processing	34
2.1.3.4	RFSS Service Capability Polling.....	35
2.1.4	<i>Service Access Point Model</i>	35
2.2	ISSI INTERNET PROTOCOL SUITE	35
2.3	ISSI DATA TRANSMISSION ORDER	35
3.	SIP MESSAGES AND PARAMETERS DEFINITION FOR ISSI	36
3.1	PREAMBLE: GRAMMAR	36
3.2	SIP TRANSPORT.....	36
3.3	REQUEST LINE CONSTRUCTION	36
3.4	SIP URI AND RELATED ADDRESSING LOGIC	36
3.4.1	<i>General Characteristics of SIP URIs</i>	36
3.4.1.1	ISSI Domain Names	36
3.4.1.2	ISSI Radical Names.....	36
3.4.1.3	ISSI Parameters.....	37
3.4.2	<i>Request URI formatting Summary</i>	37
3.5	SIP HEADER ENCODING	37
3.5.1	<i>Accept</i>	37
3.5.2	<i>Allow</i>	37
3.5.3	<i>Call-ID</i>	37
3.5.4	<i>Contact</i>	37
3.5.5	<i>Content-Disposition</i>	38
3.5.6	<i>Content-Length</i>	38
3.5.7	<i>Content-Type</i>	38
3.5.8	<i>Cseq</i>	38
3.5.9	<i>Expires</i>	38
3.5.10	<i>From</i>	38
3.5.10.1	From Header Field for REGISTER, INVITE, CANCEL, ACK, and BYE.....	38
3.5.10.2	From Header Field for OPTIONS.....	39
3.5.10.3	From Header field for MESSAGE.....	39
3.5.11	<i>Max-Forwards</i>	39
3.5.12	<i>MIME-Version</i>	39
3.5.13	<i>Priority</i>	39
3.5.14	<i>Record-Route</i>	39
3.5.15	<i>Route</i>	39
3.5.16	<i>To</i>	39
3.5.16.1	To Header Field for REGSITER	39
3.5.16.2	To Header Field for INVITE, CANCEL, ACK, and BYE	40
3.5.16.3	To Header Field for OPTIONS.....	40
3.5.16.4	To Header Field for MESSAGE.....	40
3.5.17	<i>VIA</i>	40
3.5.18	<i>Warning</i>	40
3.6	STATUS-CODE AND REASON-PHRASE	40
3.7	SDP ANNOUNCEMENTS	40
3.7.1	<i>SDP offer/answer usage</i>	40
3.7.2	<i>SDP parameters encoding</i>	40
3.7.2.1	Session-level part	41
3.7.2.1.1	The protocol version field.....	41

3.7.2.1.2	The origin field.....	41
3.7.2.1.3	The session name field.....	41
3.7.2.1.4	The time field.....	41
3.7.2.1.5	The connection field.....	41
3.7.2.2	Media description part.....	41
3.7.2.2.1	The media announcement field.....	41
3.7.2.2.2	Attributes fields.....	41
3.7.3	SDP Examples.....	42
3.8	X-TIA-P25-ISSI BODY PARAMETERS.....	42
3.8.1	Group Service Profile attributes.....	42
3.8.2	User Service Profile attributes.....	42
3.8.3	RFSS Service Capability Profile attributes.....	42
3.8.4	ISSI specific register parameter attributes.....	42
3.8.5	ISSI specific call parameter attributes.....	42
3.9	X-TIA-P25-SNDP BODY PARAMETERS.....	42
3.10	EXAMPLES.....	42
3.10.1	Mobility Management Examples [Informative].....	43
3.10.2	Call Processing Examples [Informative].....	43
3.10.3	RFSS Service Capability Polling Examples [Informative].....	43
4.	RTP MESSAGE VOCABULARY FOR ISSI.....	44
5.	MOBILITY MANAGEMENT.....	45
5.1	DEFINITION OF MOBILITY OBJECTS.....	45
5.2	VARIATIONS OF THE SIP REGISTER MESSAGE.....	45
5.3	MM-SAP/MESSAGE MODELING.....	45
5.4	STATE BASED SUMMARY.....	45
5.5	MOBILITY MANAGEMENT PROCEDURES.....	45
6.	CALL CONTROL.....	47
6.1	CC-SAP.....	47
6.2	SIP TRANSACTIONS.....	47
6.3	STATE MODELS FOR ISSI VOICE SERVICES.....	47
6.3.1	State Model: SU-to-SU Call – Originating RFSS.....	47
6.3.2	State Model: SU-to-SU Call – Destination RFSS.....	47
6.3.3	State Model: Group Call – Originating RFSS.....	47
6.3.4	State Model: Group Call – Destination RFSS.....	48
6.3.5	Sub-State Model: Group Call – RTP Session Open State.....	48
6.3.6	Sub-State Model: Group Call – No RTP Resources at Peer RFSS State.....	48
6.3.7	Sub-State Model: Group Call – No RTP Resources at Local RFSS State.....	48
6.4	BASIC ISSI SIP CONTROL PROCEDURES.....	48
6.5	CALL CONTROL PROCEDURES FOR ISSI SU-TO-SU VOICE SERVICES.....	48
6.5.1	General.....	48
6.5.2	ISSI SU-to-SU Voice Call Setup.....	48
6.5.2.1	Actions Required at Calling Serving RFSS.....	49
6.5.2.2	Actions Required at Calling Home RFSS.....	49
6.5.2.3	Actions Required at Called Home RFSS.....	49
6.5.2.4	Actions Required at Called Serving RFSS.....	49
6.5.3	ISSI Support for Roaming of the Called SU.....	49
6.5.4	ISSI Support for Roaming of the Calling SU.....	49
6.5.5	ISSI SU-to-SU Voice Call Release by Calling Serving RFSS.....	49
6.5.6	ISSI SU-to-SU Voice Call Release by Calling Home RFSS.....	49
6.5.6.1	Actions Required at Calling Home RFSS.....	49
6.5.6.2	Actions Required at Called Home RFSS.....	49
6.5.6.3	Actions Required at Called Serving RFSS.....	50
6.5.6.4	Actions Required at Calling Serving RFSS.....	50
6.5.7	ISSI SU-to-SU Voice Call Release by Called Home RFSS.....	50
6.5.8	ISSI SU-to-SU Voice Call Release by Called Serving RFSS.....	50

6.6	CALL CONTROL PROCEDURES FOR ISSI GROUP VOICE SERVICES	50
7.	PUSH-TO-TALK (PTT) MANAGEMENT	51
8.	RFSS CAPABILITY POLLING	52
	ANNEX A (NORMATIVE) GENERIC SIP RESPONSES OVER THE ISSI	53
	ANNEX B (NORMATIVE) TABLE OF TIMERS AND CONSTANTS.	54
	ANNEX C (INFORMATIVE) – TIA-102 VOCODER ERROR MITIGATION DATA.....	55
	ANNEX D (INFORMATIVE) – SU-TO-SU CALL FLOWS.....	56
	ANNEX E (INFORMATIVE) GROUP CALL FLOWS.....	57
	ANNEX F (INFORMATIVE) – SU-TO-SU CALL FLOWS INVOLVING MULTIPLE VOCODER MODES.....	58
	ANNEX G (INFORMATIVE) GROUP CALL FLOWS INVOLVING MULTIPLE VOCODER MODES.....	59
	ANNEX H (INFORMATIVE) FUTURE CONSIDERATIONS.....	60
	ANNEX I (INFORMATIVE) CONSOLE FUNCTIONALITY.....	61

List of Figures

FIGURE 1 - ABRIDGED GENERAL SYSTEM MODEL SHOWING THE IWF	22
FIGURE 2 - RFSS WITH EMBEDDED CONSOLES CONNECTS TO IWF VIA ISSI	22
FIGURE 3 - CONSOLE SUB-SYSTEM (CSS) CONNECTS TO IWF USING CSSI	23
FIGURE 4 - IWF CONNECTED TO A TIA-102 RFSS WITH ISSI AND A CSS WITH CSSI	23
FIGURE 5 - IWF CONNECTED TO A TIA-102 RFSS WITH ISSI AND TIA-102 RFSS CONNECTED TO A CSS WITH A CSSI.....	24
FIGURE 6 ADAPTATION OF [102BACA] FIGURE 9 ISSI FUNCTIONAL ENTITIES	29

Limited Use Copy

DOCUMENT REVISION HISTORY

Version	Date	Description
1	February, 2021	Approved for publication.

Limited Use Copy

A. Addendum Introduction

This Addendum to TIA-102.BACA-B, November 2012, “Project 25 Inter-RF Subsystem Interface Messages and Procedures for Voice and Mobility Management Services” [102BACA], adds an Annex which defines TIA-102 and 3GPP MCPTT interconnectivity via an Interworking Function (IWF). The interconnection between the TIA-102 system and IWF is achieved via the ISSI. See the Annex for further details.

B. Addendum Scope

This document will only include information relevant to interworking with an IWF. The reader is referred to [102BACA] and its other addendums for all other pertinent information. The next complete revision of [102BACA] will incorporate the changes contained in this addendum. Care must be taken when incorporating changes in this addendum to avoid conflicts with the changes made by other addendum that are incorporated at the same time. These include the addendum to add support for group emergency behaviors, the addendum to fix errors and omissions, and possibly others.

Changes to [102BACA] are described in this document starting at Section 1. *[Italic text between square brackets provides instructions on how to incorporate changes in a section.]* If a section contains subsections, and neither the section nor subsections thereof change, only the section will be identified in this document. Note that section numbers identified in this addendum correspond to the same section numbers in [102BACA], with new or modified text shown in orange to assist the reader.

C. Addendum References

This addendum references the following TIA documents.

C.1 Normative References

[102BACA] Project 25 *Inter-RF Subsystem Interface Messages and Procedures for Voice Services, Mobility Management, and RFSS Capability Polling Services*, TIA-102.BACA-B, TIA-102, TIA, November 2012; and Addendum B-1 (July 2013); and Addendum B-2 (November 2016)

C.2 Informative References

[TIA_STYLE] *TIA Standards Style Guide for Engineering Committees*, TIA, 2019

Limited Use Copy

1. Introduction

[No change from existing text in this section of [102BACA], except for selected subsections thereof, as shown below.]

1.1 Scope

[No change from existing text in this section of [102BACA].]

1.2 Terminology

[No change from existing text in this section of [102BACA].]

1.3 Document Structure

[No change from existing text in this section of [102BACA].]

1.4 Definitions

[No change from existing text in this section of [102BACA].]

1.5 Abbreviations

[No change from existing text in this section of [102BACA].]

1.6 References

[No change from existing text in this section of [102BACA], except for subsection 1.6.2 below.]

1.6.1 Normative References

[No change from existing text in this section of [102BACA].]

1.6.2 Informative References

[Add these references to this section of [102BACA].]

[JLMRLTE 38R002] *Liaison from JLRMLTE to TR-8, WTSC-JLMRLTE-2019-00038R002, ATIS WTSC/JLMRLTE, July 17, 2019*

[JLMRLTE 28R001] *JLMRLTE Meeting Notes May 8, ATIS WTSC/JLMRLTE, 2019 WTSC-JLMRLTE-2019-00028R01, July 17, 2019*

[JLMRLTE 01R006] *Study of interworking between P25 LMR and 3GPP mission critical service, ATIS WTSC/LMRLTE, WTSC-JLMRLTE-2019-00001R06, July 12, 2019*

[102BACF] *Project 25 Inter-RF Subsystem Interface - Messages and Procedures for Packet Data, TIA-102.BACF, TIA, October 2019*

[TR-8.19 19-023-R1] *Liaison from TR-8.19 to JLMRLTE, 19-023-R1, TR-8.19, October 4, 2019*

[JLMRLTE 11R001] *Liaison from JLRMLTE to TR-8, WTSC-JLMRLTE-2020-00011R001, ATIS WTSC/JLMRLTE, March 9, 2020*

[JLMRLTE 04R002] *Study of interworking between P25 LMR and 3GPP mission critical service – Phase 2, ATIS WTSC/LMRLTE, WTSC-JLMRLTE-2020-00004R02, March 9, 2020*

[JLMRLTE 40R000] *Liaison from JLRMLTE to TR-8, WTSC-JLMRLTE-2020-00040R000, ATIS WTSC/JLMRLTE, July 16, 2020*

[JLMRLTE 04R006] *Study of interworking between P25 LMR and 3GPP mission critical service – Interim Version, ATIS WTSC/LMRLTE, WTSC-JLMRLTE-2020-00004R006, July 16, 2020*

2. Overview of Architecture and Protocol Suite

*[No change from existing text in this section of **[102BACA]**]*

Limited Use Copy

3. SIP Messages and Parameters Definition for ISSI

*[No change from existing text in this section of **[102BACA]**]*

Limited Use Copy

4. RTP Message Vocabulary for ISSI

[No change from existing text in this section of [102BACA]].

Limited Use Copy

5. Mobility Management

*[No change from existing text in this section of **[102BACA].**]*

Limited Use Copy

6. Call Control

*[No change from existing text in this section of **[102BACA].**]*

Limited Use Copy

7. Push-to-Talk (PTT) Management

*[No change from existing text in this section of **[102BACA]**.]*

Limited Use Copy

8. RFSS Service Capability Polling

*[No change from existing text in this section of **[102BACA]**.]*

Limited Use Copy

Annex A (Normative) – Generic SIP Responses Over the ISSI

[No change from existing text in this section of [102BACA].]

Annex B (Normative) – Table of Timers and Constants

[No change from existing text in this section of [102BACA].]

Limited Use Copy

Annex C (Informative) – TIA-102 Vocoder Error Mitigation Data

[No change from existing text in this section of [102BACA].]

Limited Use Copy

Annex D (Informative) – SU-to-SU Call Flows

*[No change from existing text in this section of **[102BACA].**]*

Limited Use Copy

Annex E (Informative) – Group Call Flows

[No change from existing text in this section of [102BACA].]

Limited Use Copy

Annex F (Informative) – SU-to-SU Call Flows Involving Multiple Vocoder Modes

*[No change from existing text in this section of **[102BACA]**.]*

Limited Use Copy

Annex G (Informative) – Group Call Flows Involving Multiple Vocoder Modes

*[No change from existing text in this section of **[102BACA].**]*

Limited Use Copy

Annex H (Informative) – Future Considerations

[No change from existing text in this section of [102BACA].]

Limited Use Copy

Annex I (Informative) – Console Functionality

*[No change from existing text in this section of **[102BACA].**]*

Limited Use Copy

Annex J (Normative) – Interworking with an IWF

[Add this Annex to [102BACA].]

J.1 Introduction

This Annex defines TIA-102 and 3GPP MCPTT interconnectivity via an Interworking Function (IWF). The interconnection between the TIA-102 system and IWF is achieved via the ISSI.

J.2 Purpose and Scope of this Annex

This purpose of this Annex is to describe how an IWF may act as a Home or Serving RFSS for the following interworking services:

- Group Registration
- Group De-registration
- Group Call (non-emergency, emergency, unprotected, and protected)
- SU-to-SU call (non-emergency, emergency, unprotected, and protected)
- Announcement Group Call
- Broadcast Group Call
- Group emergency cancel
- Console participation in group call and SU-to-SU call
- Registration for group-destined Supplementary Data services

The following items are out of scope of the present Annex:

- Unit Roaming²
- Supplementary Data Services other than group emergency cancel and registration for group-destined Supplementary Data Services
- Packet Data

The structure of the IWF, and the interface between the IWF and the 3GPP MC system are out of scope of the present Annex. The definition of reference points between the IWF and TIA-102 systems and the interactions between the IWF and TIA-102 systems are in scope of the present Annex.

J.3 Architecture

² “Unit Roaming” refers to an SU or UE that has moved from its unit home system to a serving system. When this annex states that “unit roaming is not in scope” it refers to LMR SUs physically located in the 3GPP system or 3GPP UEs physically located in the LMR system; neither of these scenarios are supported and are thus “not in scope.” RFSSs can support roaming of their groups to other RFSSs; that is referred to as “group roaming.” Refer to [102BACA] for further detail regarding unit roaming and group roaming.

The Inter-RF Subsystem Interface (G Interface) is used to interconnect TIA-102³ RFSSs to create systems or Wide Area systems or Wide Area networks. The ISSI may also be used to interconnect an RFSS with an Inter-Working Function (IWF), providing connectivity to a 3GPP MCPTT system.

J.4 Overview

This section is included to inform the readers of the possible scenarios for the embedded consoles or the console subsystems to be connected with an IWF. In Figure 1 through Figure 5, the IWF-x interface is shown for reference only. IWF-x refers to the collection of interfaces between an IWF and the MC System. For further information, see [JLMRLTE 04R006].

Figure 1 shows an Abridged⁴ General System Model diagram. An RFSS is connected to an IWF via the ISSI. The IWF in turn connects to a 3GPP MCPTT System.

The present Annex describes the architecture to support the interworking between an MC system and the TIA-102 system.

From the MCPTT System perspective, the IWF, along with the TIA-102 RFSS, will appear as a peer interconnected MC system. Conversely, from the TIA-102 RFSS perspective, the IWF, along with the MC system, will appear as a peer interconnected RFSS. This Annex defines interactions on the ISSI between an IWF and a TIA-102 RFSS.

³ All instances of “RFSS” and “SU” in this Annex refer to a TIA-102 standard RFSS and TIA-102 standard SU. Similarly, “UE” refers to 3GPP Mission Critical user equipment.

⁴ Other TIA-102 interfaces omitted to enable focus on connectivity with the IWF.

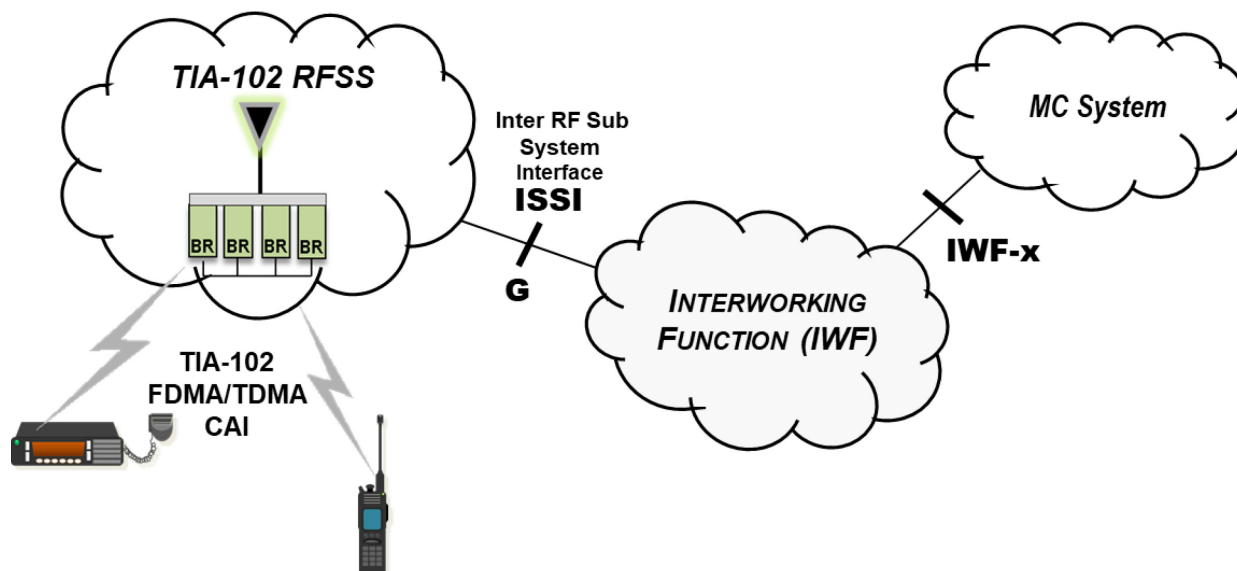


Figure 1 - Abridged General System Model Showing the IWF

J.4.1 System Diagrams with Consoles and CSSI

The following figures show how RFSSs with embedded consoles, and Console Subsystems could be connected with an IWF.

Figure 2 shows a system model where the TIA-102 RFSS has embedded consoles and connects to the IWF with the ISSI. The embedded consoles can be members of interworking groups homed to the TIA-102 RFSS (without using the ISSI). The embedded consoles can also be members of the interworking groups homed to the MC System.

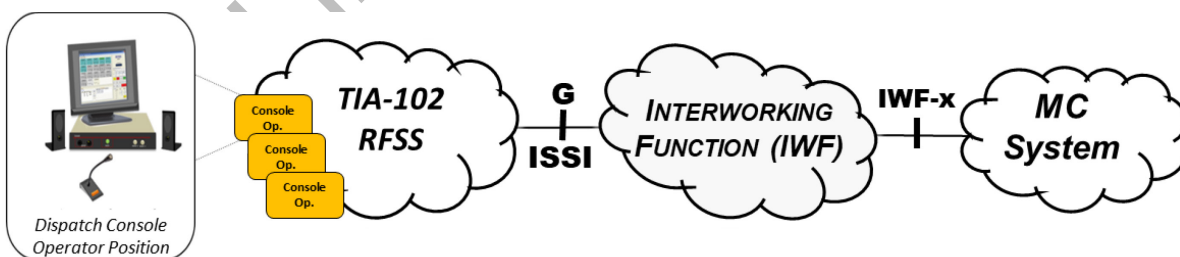


Figure 2 - RFSS with Embedded Consoles Connects to IWF via ISSI

Figure 3 shows a system model where a Console Subsystem (CSS) is connected to the IWF with the CSSI. The consoles can be members of interworking groups homed to the MC System. The CSS is not a group home and therefore is not

home to interworking groups.

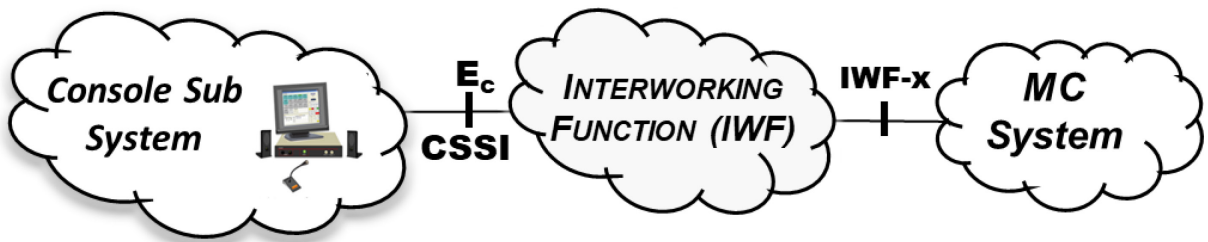


Figure 3 - Console Sub-system (CSS) connects to IWF using CSSI

Figure 4 shows a system model where a Console Subsystem (CSS) is connected to the IWF with the CSSI and the IWF also has an ISSI connection to a TIA-102 RFSS with embedded consoles. In this case the MC System is home to the interworking groups and members of both the TIA-102 RFSS and CSS can be members of those interworking groups. The next figure covers the case when the TIA-102 RFSS is the interworking group home.

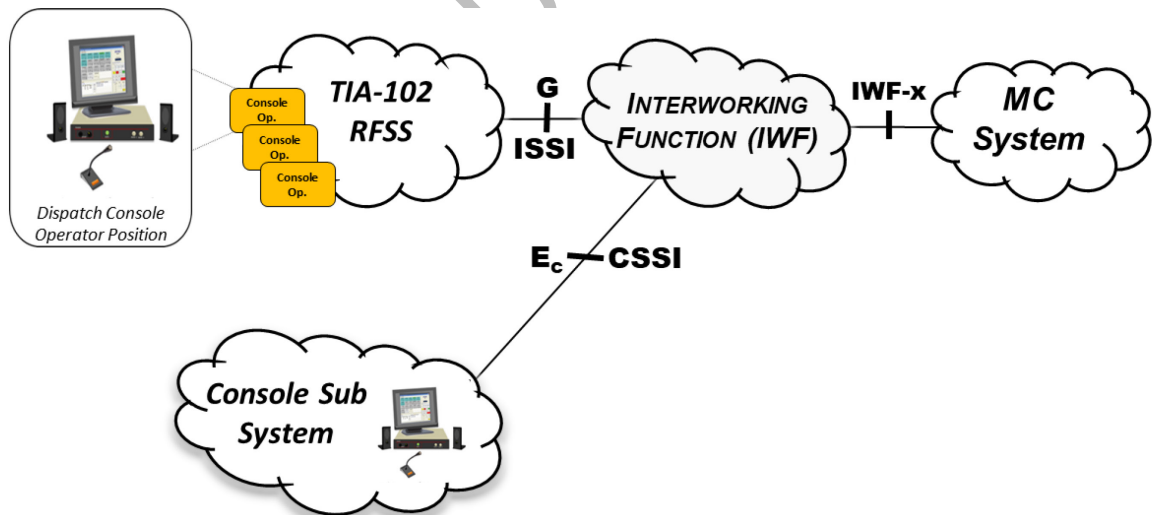


Figure 4 - IWF connected to a TIA-102 RFSS with ISSI and a CSS with CSSI

Figure 5 shows a system model where a TIA-102 RFSS has embedded consoles and connects to the IWF with the ISSI, and the TIA-102 RFSS also has a CSSI connection to a CSS. In this case the TIA-102 RFSS is home to the interworking groups and members of both the MC System and CSS can be members of the

interworking groups (note that members of the TIA-102 RFSS can also be members of the interworking groups but that does not use the ISSI or CSSI).

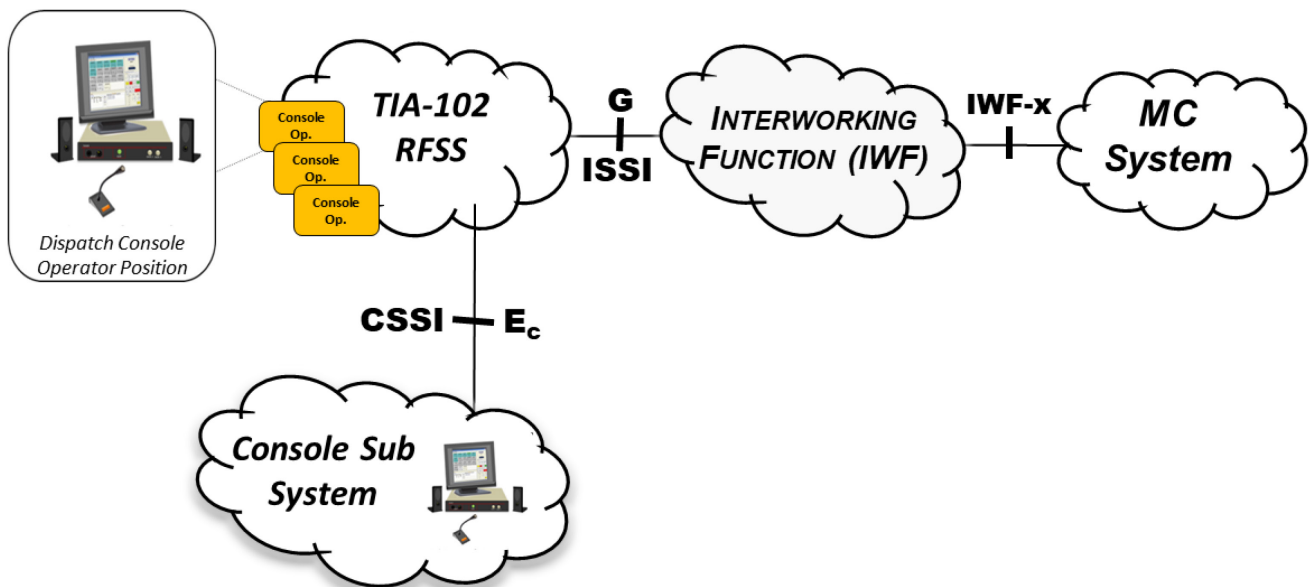


Figure 5 - IWF connected to a TIA-102 RFSS with ISSI and TIA-102 RFSS connected to a CSS with a CSSI

With a few exceptions identified in the text, the following subclauses map directly to corresponding subclauses in the body of TIA-102.BACA-B. Each subclause describes how the respective subclauses are impacted when an IWF is utilized to support interworking between a MC system and a TIA-102 system.

2. Overview of Architecture and Protocol Suite

2.1 Architecture Overview

The ISSI enables interworking between an RFSS and an IWF. SUs and RF sites in the TIA-102 system can support calls using the FDMA air interface or the TDMA air interface. Therefore, the ISSI connection from the TIA-102 RFSS to the IWF can utilize the Full Rate (FDMA) or Half Rate (TDMA) vocoder modes. A call's individual talk spurts might change from one vocoder mode to the other based on the capabilities of the SU(s) and the RF sites where they are registered, or based on the capabilities of the UEs and LTE equipment in the MC System. On the ISSI the vocoder mode is negotiated on a talk spurt by talk spurt basis.

NOTE: Vocoder mode negotiation where a vocoder mode of Legacy (IMBE), Full Rate, Half Rate, or Native is chosen for a connection, and the fact that negotiation can occur on a talk-spurt by talk-spurt basis are areas that need input from JLMRLTE. Input is needed because it is not known if there will be limitations imposed when an RFSS is an IWF. Also, transcoding and trans encrypting should be studied for the case of MCPTT UEs using AMR vocoders while TIA SUs use narrowband AMBE vocoders.

The ISSI standard uses the term SU to denote either a console or a mobile radio. The ISSI defines a Console as a type of SU (denoted cSU). In a MC System, LTE UEs are not differentiated between subscribers and consoles except via configuration of special privileges for LTE console UEs. An IWF connected to an LMR system proxies for the LTE UEs and is expected to map LTE UE's to LMR SUs. Likewise, an IWF will map an interworking LTE talkgroup identity to a TIA-102 talkgroup identity that is used when interacting with an RFSS.

NOTE: The differences between LMR consoles and mobile SUs, and how that might translate to the MC system side is something JLMRLTE should study.

As an RFSS connected via the ISSI, the IWF takes on the role of an individual Home RFSS for its interworking UEs and may take on the role of the Home RFSS for interworking groups that are homed in the 3GPP MC system. The IWF can also take on the role of a Serving RFSS for interworking groups that are homed in the RFSS. The IWF will not take on the role of a Serving RFSS for a unit that is homed in the RFSS because unit registration for interworking UEs and TIA-102 SUs is out of scope of the present Annex.

As a Home RFSS for an interworking group, the IWF negotiates the vocoder mode on each call segment with Serving RFSSs. The IWF can be a Home

RFSS for its own interworking groups and a Serving RFSS for interworking groups homed in the TIA-102 system at the same time.

When connected to an LMR RFSS, and because unit roaming is not in scope of the present annex, for an interworked SU-to-SU call the IWF can take on either the Called Home RFSS role or the Calling Home RFSS role.

When taking on the Called Home role for a UE participating in an interworked SU-to-SU call, the IWF chooses the initial transmitter (calling SU or called SU) for the SU-to-SU call and informs the Calling Home (the LMR RFSS) of its choice. As Called Home the IWF also determines whether the SU-to-SU call employs availability check or direct mode with the called SU.

The ISSI employs the SIP and RTP protocols on top of UDP and IP. Unlike MCPTT, the ISSI does not use RTCP for transmission control (RTCP is not utilized at all). On the ISSI, transmission control and the conveyance of voice both utilize RTP.

2.1.1 Call Models

The IWF participates in group calls and SU-to-SU calls as an RFSS connected via the ISSI.

The ISSI call model **[102BACA]** Figure 4 applies to group calls and SU-to-SU calls between the IWF and TIA-102 RFSS. The ISSI hang time is independent of the RF hang time on the TIA-102 side.

The general ISSI call model depicted in **[102BACA]** Figure 8 applies for group calls and SU-to-SU calls. Because of losing audio, interworked calls that the IWF participates in might have zero, one or more active talk spurts in each direction at any particular time.

2.1.1.1 Losing Audio Scenarios

New Clause: Interworked group calls that the IWF participates in can be transmission trunked or message trunked on the TIA-102 side as depicted in **[102BACA]** Figure 3. The overlap of two transmissions at the same time on the TIA-102 side can create “losing audio” (the audio not chosen to be repeated) which will be identified as such to the IWF. Thus, the IWF might receive two or more streams of audio (all but one marked as “losing”) for a call originating on the TIA-102 side. The TIA-102 side can accept “losing audio” from the IWF as well. Note that the standard allows an RFSS to mark the only audio stream associated with a call as “losing”.

2.1.1.2 Confirmed and Unconfirmed Group Calls

New Clause: Group calls on the ISSI can be either “confirmed” or “unconfirmed”. The group home RFSS configuration drives whether a group uses confirmed or unconfirmed group call. When the IWF is the group home of the interworking group its own configuration will determine whether its groups operate in confirmed or unconfirmed mode. When the TIA-102 system is the group home (and the IWF is a group serving) for a group using confirmed group call, the IWF is expected to indicate to the group home (in its requests/responses) whether it has resources. When a confirmed call is setup the initial talker is queued until all the required resources are available, or a timeout occurs. An unconfirmed call makes no effort to guarantee participation of particular RFSSs. The call starts upon the beginning of the first transmission.

2.1.2 Subsystem Topologies

The typical group call interconnection model shown in **[102BACA]** Figure 5 applies except an IWF will not be connected directly to another IWF via an ISSI. When the IWF is a Serving RFSS relative to the interworking group, it proxies for all LTE UEs group members operating in the MCPTT System. When the IWF is a Home RFSS relative to the interworking group it will fan out the audio to the other interested Serving RFSSs.

NOTE: Figure 5 depicts a Home RFSS connected to three distinct Serving RFSSs. This is a typical architecture used to connect different LMR RFSSs. The IWF can take on the Home RFSS role for an interworking group and a scenario such as that shown in Figure 5 with more than one LMR Serving system is not currently depicted in the JLMRLTE Study.

When the IWF is the Home RFSS relative to the interworking group it will also choose the allowed vocoder modes for the talk spurts on each particular call segment based on its needs and the desired vocoder modes of the interested Serving RFSSs. When the IWF is a Serving RFSS relative to the interworking group, the IWF negotiates vocoder mode with the Home as if the IWF were a Serving RFSS. If the Home and Serving RFSS agree upon Native mode, then the vocoder mode of the talk spurts on the associated ISSI call segment might change from one talk spurt to another.

The typical unit-to-unit call interconnection model shown in **[102BACA]** Figure 6 applies except the call segment between an IWF RFSS and an LMR RFSS is always the one between a calling home RFSS and a called home RFSS (each RFSS can be one or the other). This is because unit roaming is not in scope of the present Annex. Figure 7 Unit-to-Unit Call after In-call Roaming in **[102BACA]** is also not in applicable because unit roaming is not in scope of the present Annex.

2.1.3 Functional Model

Mobility Management, Call Control, Media Processing (Transmission Control) and RFSS Service Capability Polling all apply in whole or in part as described below.

Mobility Management is a set of functional services corresponding to the affiliation and authentication functions of the TIA-102 air interface. Note that in this case it does not include registration. For Mobility Management, the IWF takes on the Group Mobility Management Function (GMMF) and the Unit Mobility Management Function (UMMF) but not the Packet Data Mobility Management Function (PDMMF). Neither the PDMMF, nor Packet Data in general are in scope of this annex, see **[102BACF]**.

Call Control is a set of functional services related to control signaling for ISSI calls. It includes mechanisms for:

- call setup,
- call teardown,
- management of RTP resources,
- determination of the allowed vocoder mode for talk spurts on an ISSI call segment,
- the coordination of confirmed calls
- and the coordination of Group Emergency Cancelation (**[102BACA]** Addendum B-1).

For Call Control the IWF takes on the Group Call Control Function (GCCF) and the Unit Call Control Function (UCCF).

Media Processing (Transmission Control) is concerned with the management of talk spurts within an ISSI call. It includes mechanisms for arbitrating simultaneous call requests, for the management of losing audio, the conveyance of voice information and arbitrating between multiple console requests.

For Media Processing the IWF takes on both the Master Media Function (MMF) and the Subordinate Media Function (SMF). Which media function is active for a particular call is determined by the role of the IWF in the Call (Home or Serving). Media processing is provided via RTP.

For RFSS Service Capability Polling the IWF takes on the RFSS Service Capability Polling Function (RCPF). Although RFSS Service Capability Polling is an optional support function, the IWF is required to respond to a polling request (SIP OPTIONS).

Below is an adaptation from Figure 9 of **[102BACA]** and Figure 11 of **[102BACF]** showing the functional entities that are not present when connecting to an IWF for MCPTT interworking.

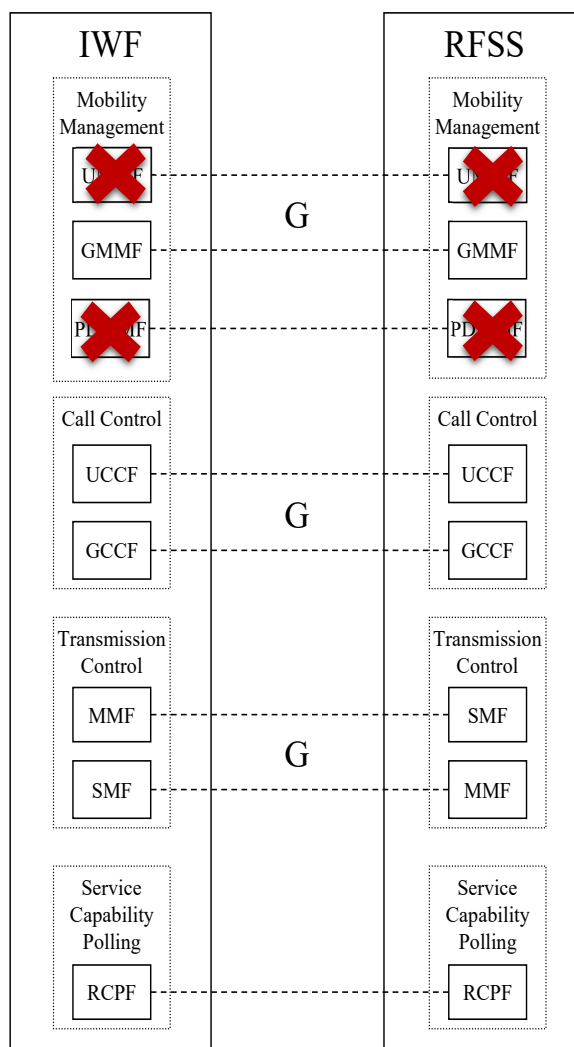


Figure 6 Adaptation of [102BACA] Figure 9 ISSI Functional Entities

2.1.3.1 Mobility Management

As stated previously, Unit Mobility Management does not apply because unit registration is not a supported service for interworking.

The Group Mobility Management services group affiliation, group tracking, and group supplementary data interest management apply.

Group Supplementary Data Monitoring Interest management does not apply because supplementary data monitoring interest is out of scope of the present Annex.

When an IWF is a Serving RFSS relative to the group, its GMMF is responsible for:

- Registration of the group with the Home RFSS when the RFSS detects it has SUs currently affiliated with the group in its serving area (i.e. the MCPTT System),
- As part of registration reporting to the Home RFSS its supported ISSI vocoder mode(s) for the group listed in preferred order (possible modes include Full Rate, Half Rate, and Native mode).
- De-registration of the group or announcement subscriber group with the Home RFSS when the RFSS determines that it no longer has affiliated members for the group,
- Responding to the Home RFSS when the Home RFSS requests confirmation that the RFSS still has an interest in being registered for the group,
- Requesting service profile information for the group on the ISSI only when the RFSS is currently registered to the SG or is in the process of registering with the group.
- Registration of the announcement subscriber group associated with a group that the Serving RFSS has registered if it has not previously registered that announcement subscriber group.
- Registering interest in Supplementary Data for Group (becoming a Group Serving RFSS).

When the IWF is the Home RFSS relative to the group, its GMMF is responsible for the following functions:

- Providing the group service profile information to suitably authorized Serving RFSSs that request the information.
- Providing its supported vocoder modes listed in preferred order (possible modes include Full Rate, Half Rate, and Native mode) for the SG to suitably authorized Serving RFSSs that request the information.
- Prohibiting registration of the group in serving domains for which it is not authorized.
- Maintaining a list of Serving RFSSs of the SG and their supported vocoder modes (possible modes include Full Rate, Half Rate, and Native mode), if known.
- Checking whether a Serving RFSS still has an interest in being registered for the SG, in accordance with local policy.
- Maintaining a distribution list of Group Serving RFSSs with interest in Supplementary Data for each group homed to the RFSS.

Mobility Management for Supplementary Data applies for the case when the supplementary data messages are address to a Group. The case where supplementary data messages are addressed to an individual SU or to General FNE are out of scope of the present Annex. Monitoring of supplementary data transactions that involve an SU, a Group or General FNE is also out of scope of the present Annex.

Packet Data Mobility Management does not apply because Packet Data is out of scope of the present Annex.

2.1.3.2 Call Control

2.1.3.2.1 Unit Call Control

Unit Call Control applies and comprises the following services:

- Negotiation of SU-to-SU call type (with or without the optional availability check)
- Negotiation of the vocoder mode that can be used with the talk spurts on each SU-to-SU ISSI call segment
- Setup of SU-to-SU calls that employ the optional availability check
- Setup of SU-to-SU calls that do not employ the optional availability check
- SU-to-SU call termination
- Identification of the Initial Talking Radio

Note that Unit Call Control does not include call transfer during SU roaming because unit roaming is not in scope of the present Annex.

The IWF does not take on the role of the Calling RFSS or Called RFSS because unit roaming is not in scope of the present Annex.

When the IWF is a Calling Home RFSS, its UCCF is responsible for:

- Propagation of an SU-to-SU call set up to the Called Home RFSS,
- Participation in the determination of the type of unit-to-unit call (with or without availability check),
- Participation in the determination of the allowed ISSI vocoder mode for the talk spurts on the call segment to the Called Home RFSS,
- Propagation of call preferences,
- Creating an RTP session with the Called Home RFSS,
- Propagation of the call release to the Called Home RFSS,
- Releasing of the call on loss of RTP connectivity, on RTP resource preemption or other local policy.

When the IWF is a Called Home RFSS, its UCCF is responsible for:

- Determination of the type of unit-to-unit call (with or without availability check),
- Determining the transmitting SU and holding off the start of the call until RF resources are available on both systems,

- Participation in the determination of the allowed ISSI vocoder mode for the talk spurts on the call segments to the Calling Home RFSS,
- Propagation of session progress information (such as ringing, queued and grant information) to the Calling Home RFSS,
- Creating an RTP session with the Calling Home,
- Release of the call when requested by the Calling Home RFSS,
- Release of the call on loss of RTP connectivity, on RTP resource preemption or other local policy.

In Figure 11 – Unit-to-unit Call Media Tree Topology **[102BACA]** the RTP Unicast Session between the Calling Home and Called Home is applicable when ISSI is being used to interface with an IWF.

2.1.3.2.2 Group Call Control

Group Call Control applies and comprises the following services:

- Group call set up
- Group call segment set up, including determining the allowed vocoder mode for a call segment's talk spurts
- Group call segment termination
- Group call termination
- Confirmed Call Management
- Coordination of Group Emergency Cancellation (**[102BACA]** Addendum B-1)

When the IWF is a Serving RFSS, its GCCF is responsible for:

- Requesting a group call set up when an SU triggers a group call request in its area for a group for which no call currently exists
- Informing the Home RFSS of its offered vocoder mode(s) when it requests that a group call be set up
- Participating in the establishment of new call segments in response to requests from the Home RFSS
- Releasing a group call segment with the Home RFSS on loss of RTP connectivity or other local policy condition
- Once a group call segment has been set up, tearing down or setting up RTP sessions upon preemption or Home RFSS request.
- Communicating to the Home RFSS the RTP session parameters
- Notifying the Home RFSS of the availability or lack of availability of RF resources required for a confirmed call
- Requesting a change of group call priority based on local policy
- Requesting a change of its call segment's allowed vocoder mode based on local policy
- Signaling a Group Emergency Cancellation to the Home RFSS when a group emergency is cancelled by a sufficiently privileged SU or due to local policy, such as when a Dispatch Console within the Serving RFSS cancels the group emergency.

When the IWF is a Home RFSS relative to the group, its GCCF is responsible for:

- Establishing group call segments in response to group call setup requests from Serving RFSSs or SU PTT triggers within its own area, including:
 - to get the list of registered Serving RFSSs from the GMMF
 - Collaborating with the registered Serving RFSSs to determine the allowed vocoder mode for the talk spurts on each call segment
 - to initiate group call set ups with the registered Serving RFSSs
 - to bridge RTP sessions which are set up with the Serving RFSSs
 - to handle INVITE and re-INVITE request collisions
- On-going management of group call segments during a call, including
 - To initiate group call segment set ups with any registered Serving RFSSs not already joined to the call (late call entry)
 - To inform a registered Serving RFSS of a requested change of the allowed vocoder modes for the talk spurts on the call segment, based on the Serving RFSS's request and/or based on local policy
 - To handle a group call segment set up when it receives a group call set up request from any Serving RFSS
 - Once group call segment has been set up, tearing down or setting up RTP sessions upon preemption or Serving RFSS request.
- Releasing the group call on ISSI hangtime⁵ expiry or other local policy condition
- Releasing a group call segment with a Serving RFSS on loss of RTP connectivity or other local policy condition
- Communicating to each Serving RFSS the RTP session parameters associated with its call segment
- Managing the establishment and maintenance of confirmed group calls
- Changing the group call priority upon Serving RFSS request or upon its own local policy
- Signaling a Group Emergency Cancellation to the Serving RFSSs when it receives a Group Emergency Cancel from a sufficiently privileged SU or a Serving RFSS, or when it determines the group emergency to be over through its own local policy (e.g. when a Dispatch Console internal to the Home RFSS cancels the group emergency).

When the IWF is a Home RFSS relative to the group, and if its interworked groups are confirmed, then it is informed of RF resource availability in the Serving RFSS and starts the call when the requisite resources are available. For unconfirmed call no consideration of RF resource availability is required.

⁵ Hangtime on the ISSI is not correlated to the RF resource hangtime. Generally, hangtime on the ISSI is longer than RF resource hangtime.

The GCCF is responsible for the selection of the allowed vocoder mode within the context of the talk spurts on each call segment that comprise the group call. The allowed vocoder mode for each call segment can be negotiated independently of the other call segments.

When the GCCF in the Home RFSS receives a Group Emergency Cancel from a sufficiently privileged SU or from a Serving RFSS, the GCCF signals a Group Emergency Cancel to the Serving RFSSs. Also, the GCCF could signal a Group Emergency Cancel to the Serving RFSSs based on its local policy, such as when it receives a group emergency cancel from an embedded console.

When the GCCF in the Serving RFSS receives a Group Emergency Cancel from a sufficiently privileged SU, the GCCF signals a Group Emergency Cancel to the Home RFSS. Also, the GCCF could signal a Group Emergency Cancel to the Home RFSS based on its local policy, such as when it receives a group emergency cancel from an embedded console.

2.1.3.3 Media Processing

Media Processing is a functional service which is responsible for:

- Push-To-Talk (PTT) management of competing talk spurts, and
- Conveyance of Full Rate, Half Rate, or Native mode voice information and other signaling between RFSSs participating in a call.

Media Processing applies for group calls and SU-to-SU calls. Both the Master Media Function (MMF) and the Subordinate Media Function (SMF) apply for group calls. The Home RFSS of the group provides the MMF and the Serving RFSSs employ SMFs.

For an IWF, the Master Media Function (MMF) is responsible for:

- Arbitration of competing talk spurts coming from different SMFs, including arbitrating between a console and mobiles and between multiple consoles.
- Fanout (bridging) of RTP sessions between SMFs, including copying RTP packets from a source SMF to all other SMFs participating in the call.
- Fanout of preempted talk spurts to all SMFs interested in receiving losing audio (for Group Calls).
- Monitoring of RTP session connectivity and reporting to the GCCF or UCCF in case of loss of connectivity.
- Monitoring of voice payloads and reporting to the GCCF or UCCF for ISSI hang-time management. In SU-to-SU calls, this function only applies for the
- Called Home RFSS.

For an IWF, the SMF is responsible for:

- Arbitration of competing talk spurts within its serving area, including arbitrating between a console and mobiles and between multiple consoles.

- Propagation of PTT Transmit Requests, voice activity and progress to the MMF
- Management of MMF responses to arbitration requests
- Fanout of voice activity within its serving area
- Monitoring RTP session connectivity and reporting to the GCCF or UCCF in case of loss of connectivity.

2.1.3.4 RFSS Service Capability Polling

The RFSS Serving Capability Polling is optional, but all RFSSs, including the IWF, are required to respond if they receive a request. A response can optionally carry more detailed service capability information.

2.1.4 Service Access Point Model

The Service Access Point Model applies in the following manner:

- Mobility Management SAP applies for group call / group affiliation, the “Group for Voice” mobility object and the “Group Supplementary Data” mobility object only.
- The Call Control SAP applies for group call, SU-to-SU call, and group emergency cancel ([102BACA] Addendum B-1).
- The Transmission Control SAP applies to group call and unit-to-unit call.
- The RFSS Service Capability Polling SAP applies.

2.2 ISSI Internet Protocol Suite

This section applies as written when the ISSI is being used to interface with an IWF, including the use of IPV v4 and the DSCP and CU header fields.

2.3 ISSI Data Transmission Order

This section applies as written when the ISSI is being used to interface with an IWF.

3. SIP Messages and Parameters Definition for ISSI

3.1 Preamble: Grammar

This section applies as written when the ISSI is being used to interface with an IWF.

3.2 SIP Transport

This section applies as written when the ISSI is being used to interface with an IWF.

3.3 Request Line Construction

This section applies as written when the ISSI is being used to interface with an IWF.

3.4 SIP URI and related addressing logic

3.4.1 General Characteristics of SIP URIs

This section applies as written when the ISSI is being used to interface with an IWF.

3.4.1.1 ISSI Domain Names

This section applies when the ISSI is being used to interface with an IWF.

3.4.1.2 ISSI Radical Names

This section applies when the ISSI is being used to interface with an IWF except that TIA-P25-SDMsap, snc-id, and nsapi-id are not supported because they are out of scope of the present Annex.

The contents of **[102BACA]** Table 5 ISSI Radical Names for Group Call Control Procedures applies when the ISSI is being used to interface with an IWF.

The content of **[102BACA]** Table 6 ISSI Radical Names for Unit-to-Unit Call Control Procedures applies when the ISSI is being used to interface with an IWF.

Only the Group Identifier row (sg-id) of **[102BACA]** Table 7 ISSI Radical Names for mobility management procedures applies when the ISSI is being used to interface with an IWF because unit registration, and SN registration are out of scope of the present Annex.

[102BACA] Table 8 ISSI Radical Names for RFSS Service Capability Polling Procedures applies when the ISSI is being used to interface with an IWF.

3.4.1.3 ISSI Parameters

This section applies when the ISSI is being used to interface with an IWF, but with regards to the numplan-param, only numplan-param “;user=TIA-P25-SU”, “;user=TIA-P25-SG”, “;user=TIA-P25-SD-SG”, and “;user=TIA-P25-RFSS” apply. The other numplan-params are out of scope of the present Annex. Note that later on TIA-102.BACA-B contains errors in the group call examples which incorrectly use numplan-param “;user=TIA-P25-SU.

3.4.2 Request URI formatting Summary

[102BACA] Table 10 - Summary of Request URI formats for various SIP Requests for an SU-to-SU Call applies when the ISSI is being used to interface with an.

[102BACA] Table 11 - Summary of Request URI formats for various SIP Requests for a group call and **[102BACA]** Table 12 - Summary of Request URI formats for RFSS Service Capability Polling both apply when the ISSI is being used to interface with an IWF.

3.5 SIP Header Encoding

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.1 Accept

This section applies when the ISSI is being used to interface with an IWF except for the sndcp body because packet data is out of scope of the present Annex.

3.5.2 Allow

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.3 Call-ID

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.4 Contact

This section applies when the ISSI is being used to interface with an IWF except for snc-id packet data is not in scope of the present Annex.

In [102BACA] Table 14 – Contact header field for applies except for the in-call roaming examples when the ISSI is being used to interface with an IWF.

3.5.5 Content-Disposition

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.6 Content-Length

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.7 Content-Type

This section applies when the ISSI is being used to interface with an IWF except that the x-tia-p25-sndcp body type is not supported because packet data is not in scope of the present Annex.

In [102BACA] Table 15 - Content header field used for ISSI only the rows with P25-ISSI body or an SDP body apply when the ISSI is being used to interface with an IWF.

3.5.8 Cseq

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.9 Expires

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.10 From

This section applies when the ISSI is being used to interface with an IWF except for ident equal to snc-id because packet data is not in scope of the present Annex.

3.5.10.1 From Header Field for REGISTER, INVITE, CANCEL, ACK, and BYE

This section applies when the ISSI is being used to interface with an IWF.

3.5.10.2 From Header Field for OPTIONS

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.10.3 From Header field for MESSAGE

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.11 Max-Forwards

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.12 MIME-Version

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.13 Priority

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.14 Record-Route

This section applies when the ISSI is being used to interface with an IWF.

3.5.15 Route

This section applies when the ISSI is being used to interface with an IWF.

3.5.16 To

This section applies for ident equal to sui-id, sg-id and TIA-P25-RFSSCapability only when the ISSI is being used to interface with an IWF.

3.5.16.1 To Header Field for REGSITER

This section (and its subsections) only apply for SGID registration and Group Supplementary Data registration when the ISSI is being used to interface with an IWF. SUID registration, the remaining types of supplementary data registration,

and packet data registration do not apply because they are not in scope of the present Annex.

3.5.16.2 To Header Field for INVITE, CANCEL, ACK, and BYE

This section applies when the ISSI is being used to interface with an IWF.

3.5.16.3 To Header Field for OPTIONS

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.16.4 To Header Field for MESSAGE

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.17 VIA

This section applies as written when the ISSI is being used to interface with an IWF.

3.5.18 Warning

This section applies for group call, unit-to-unit call, and group registration when the ISSI is being used to interface with an IWF.

3.6 Status-Code and Reason-Phrase

This section applies as written when the ISSI is being used to interface with an IWF.

3.7 SDP Announcements

3.7.1 SDP offer/answer usage

This section applies for group call and SU-to-SU call when the ISSI is being used to interface with an IWF. The IWF's negotiation will reflect its capabilities and that of the MC system.

[102BACA] Table 25 – SDP offer / answer summary applies when the ISSI is being used to interface with an IWF.

3.7.2 SDP parameters encoding

This section applies as written when the ISSI is being used to interface with an IWF.

3.7.2.1 Session-level part

This section applies as written when the ISSI is being used to interface with an IWF.

3.7.2.1.1 The protocol version field

This section applies as written when the ISSI is being used to interface with an IWF.

3.7.2.1.2 The origin field

This section applies as written when the ISSI is being used to interface with an IWF.

3.7.2.1.3 The session name field

This section applies when the ISSI is being used to interface with an IWF.

3.7.2.1.4 The time field

This section applies as written when the ISSI is being used to interface with an IWF.

3.7.2.1.5 The connection field

This section applies as written when the ISSI is being used to interface with an IWF.

3.7.2.2 Media description part

3.7.2.2.1 The media announcement field

This section applies as written when the ISSI is being used to interface with an IWF. The IWF's use of this field will reflect its capabilities and that of the MC system.

3.7.2.2.2 Attributes fields

This section applies as written when the ISSI is being used to interface with an IWF. The IWF's use of this field will reflect its capabilities and that of the MC system.

3.7.3 SDP Examples

This section and subsections apply as written when the ISSI is being used to interface with an IWF. The IWF's ability to perform the examples will reflect its capabilities and that of the MC system.

3.8 X-TIA-P25-ISSI Body Parameters

3.8.1 Group Service Profile attributes

This section applies when the ISSI is being used to interface with an IWF.

Note that all Group Service Profile attributes are optional and the default Group Call Type (when it is not present) is unconfirmed.

3.8.2 User Service Profile attributes

This section applies when the ISSI is being used to interface with an IWF. Note that all Unit Service Profile attributes are optional.

3.8.3 RFSS Service Capability Profile attributes

This section applies as written when the ISSI is being used to interface with an IWF. Note that all RFSS Service Capability profile attributes are optional.

3.8.4 ISSI specific register parameter attributes

This section applies as written when the ISSI is being used to interface with an IWF. Note that all ISSI Specific Register Parameter attributes are optional.

3.8.5 ISSI specific call parameter attributes

This section applies as written when the ISSI is being used to interface with an IWF. Note that all ISSI Specific Call Parameter attributes are optional.

3.9 X-TIA-P25-SNDP Body Parameters

This section does not apply when the ISSI is being used to interface with an IWF because packet data is not in scope of the present Annex.

3.10 Examples

3.10.1 Mobility Management Examples [Informative]

Only Group SD Registration (3.10.1.3) applies when the ISSI is being used to interface with an IWF. The remaining examples do not apply because they are for features and functions that are not in scope of the present Annex.

3.10.2 Call Processing Examples [Informative]

Example 1 SU Registration addressing example does not apply when the ISSI is being used to interface with an IWF because unit registration is not in scope of the present Annex.

Example 2 SU-to-SU Call addressing example does apply when the ISSI is being used to interface with an IWF.

Example 3 Unconfirmed Group Call addressing example applies when the ISSI is being used to interface with an IWF. The IWF's ability to perform this example will depend on its capability and the capabilities of the MC system.

3.10.3 RFSS Service Capability Polling Examples [Informative]

This section applies as written when the ISSI is being used to interface with an IWF. But note that the minimum requirement is to simply respond to the SIP OPTIONS polling request. Transfer of capabilities is not a requirement.

4. RTP Message Vocabulary for ISSI

This entire section applies as written when the ISSI is being used to interface with an IWF.

Note that which vocoder modes the IWF supports will reflect its capabilities and the capabilities of the MC system.

Limited Use Copy

5. Mobility Management

When the ISSI is being used to interface with an IWF, this clause applies to voice group mobility and group addressed supplementary data. Mobility of units, and the remaining mobility objects related to supplementary data and packet data are not in scope of the present annex. A brief description for how each second-tier sub-clause applies is provided.

5.1 Definition of Mobility Objects

The following two mobility objects apply when the ISSI is being used to interface with an IWF:

- The group mobility object identified by the “TIA-P25-SG” *numplan-param*. In [102BACA] Table 40 RFSS Roles, this is the group mobility object (Class 2) identified by the SG Home RFSS role and the Called Serving role. It is also termed “group addressed voice” later in the document.
- The group supplementary data mobility object identified by the “TIA-P25-SD-SG” *numplan-param*. In [102BACA] Table 40 RFSS Roles, this is the group supplementary data mobility object (Class 4) identified by the SG Home RFSS role and the Group Serving role.

5.2 Variations of the SIP Register message

The three variations of the SIP REGISTER message described in this section apply for the group mobility object (Class 2) and the group supplementary data mobility object (Class 4) when the ISSI is being used to interface with an IWF.

5.3 MM-SAP/Message Modeling

The four procedures in [102BACA] Figure 60 MM SAP/Message Model apply for the group mobility object (Class 2) and the group supplementary data mobility object (Class 4) when the ISSI is being used to interface with an IWF.

5.4 State Based Summary

This section, including [102BACA] Figure 61 Serving Mobility Function State Model applies for the group mobility object (Class 2) and for the group supplementary data mobility object (Class 4) when the ISSI is being used to interface with an IWF.

5.5 Mobility Management Procedures

When the ISSI is being used to interface with an IWF the five mobility management procedures apply as follows:

1. Serving Registration – The Serving Registration procedures apply for the group mobility object (Class 2) and the group supplementary data mobility object (Class 4).
2. Serving Query – The Serving Query procedures do not apply because SU mobility objects and packet data are not in scope of the present Annex.
3. Home Query – The Home Query procedures apply only when the Home RFSS is querying a Serving RFSS to determine whether it still has interest in a group mobility object (Class 2) or a group supplementary data mobility object (Class 4).
4. Serving Deregistration – The Serving Deregistration procedures apply only when the Serving RFSS is removing interest in a group mobility object (Class 2) or a group supplementary data mobility object (Class 4).
5. Home Deregistration – The Home Deregistration procedures apply only when the Home RFSS is deregistering a group mobility object (Class 2) or a group supplementary data mobility object (Class 4) from a Serving RFSS.

The ten “general requirements” described in **[102BACA]** apply to these procedures except the Serving Query procedure when the ISSI is being used to interface with an IWF.

6. Call Control

Call control is applicable to group call and SU-to-SU call when the ISSI is being used to interface with an IWF.

6.1 CC-SAP

The Call Control Service Access Point (CC-SAP) applies to group call and SU-to-SU call when the ISSI is being used to interface with an IWF.

Both the SU-to-SU Call and the Group Call column of **[102BACA]** Table 43 – Parameters of the CC-SAP Primitives applies when the ISSI is being used to interface with an IWF.

6.2 SIP Transactions

This section applies when the ISSI is being used to interface with an IWF.

6.3 State Models for ISSI Voice Services

The SU-to-SU call and the group call state models apply when the ISSI is being used to interface with an IWF. In **[102BACA]** Figure 64 SU-to-SU RFSS Roles the SU-to-SU Call Setup applies while the SU-to-SU In-Call Roaming does not apply because unit roaming is not in scope of the present Annex. **[102BACA]** Figure 65 Group Call RFSS Roles applies as well.

Note that for SU-to-SU Call Setup only the interface between the Calling Home RFSS and the Called Home RFSS applies when the ISSI is used to interface with an IWF. This is because unit roaming is not in scope of the present Annex.

6.3.1 State Model: SU-to-SU Call – Originating RFSS

When the ISSI is being used to interface with an IWF this sub-clause applies when the Originating RFSS is the Calling Home RFSS.

6.3.2 State Model: SU-to-SU Call – Destination RFSS

When the ISSI is being used to interface with an IWF this sub-clause applies when the Destination RFSS is the Called Home RFSS.

6.3.3 State Model: Group Call – Originating RFSS

This sub-clause applies when the ISSI is being used to interface with an IWF.

6.3.4 State Model: Group Call – Destination RFSS

This sub-clause applies when the ISSI is being used to interface with an IWF.

6.3.5 Sub-State Model: Group Call – RTP Session Open State

This sub-clause applies when the ISSI is being used to interface with an IWF.

6.3.6 Sub-State Model: Group Call – No RTP Resources at Peer RFSS State

This sub-clause applies when the ISSI is being used to interface with an IWF.

6.3.7 Sub-State Model: Group Call – No RTP Resources at Local RFSS State

This sub-clause applies when the ISSI is being used to interface with an IWF.

6.4 Basic ISSI SIP Control Procedures

The basic ISSI SIP control procedures in this clause apply when the ISSI is being used to interface with an IWF.

6.5 Call Control Procedures for ISSI SU-to-SU Voice Services

These seven types of call control procedures are listed in this sub-clause:

1. ISSI SU-to-SU Voice Call Setup
2. ISSI Support for Roaming of the Called SU
3. ISSI Support for Roaming of the Calling SU
4. ISSI SU-to-SU Voice Call Release by Calling Serving RFSS
5. ISSI SU-to-SU Voice Call Release by Calling Home RFSS
6. ISSI SU-to-SU Voice Call Release by Called Home RFSS
7. ISSI SU-to-SU Voice Call Release by Called Serving RFSS

Items 1, 5, and 6 apply when the ISSI is used to interface to an IWF. The remaining items (2,3,4, and 7) are not applicable because unit roaming is not in scope of the present Annex.

6.5.1 General

When the ISSI is used to interface to an IWF this section applies for the call segment between the Calling Home RFSS and the Called Home RFSS. The other two call segments (between the Calling Serving RFSS and Calling Home RFSS, and between the Called Home RFSS and Called Serving RFSS) are not in scope of the present Annex because unit roaming is not in scope.

6.5.2 ISSI SU-to-SU Voice Call Setup

6.5.2.1 Actions Required at Calling Serving RFSS

When the ISSI is used to interface to an IWF this sub-clause does not apply because unit roaming is not in scope of the present Annex.

6.5.2.2 Actions Required at Calling Home RFSS

When the ISSI is used to interface to the IWF, this section applies as if the IWF is both the Calling Home and Calling Serving. How the IWF interface to the 3GPP MC system is translated to its ISSI interface to the Called Home RFSS is not in scope of this Annex.

6.5.2.3 Actions Required at Called Home RFSS

When the ISSI is used to interface to the IWF, this section applies as if the IWF is both the Called Home and Called Serving. How the IWF interface to the 3GPP MC system is translated to its ISSI interface to the Calling Home RFSS is not in scope of this Annex.

6.5.2.4 Actions Required at Called Serving RFSS

When the ISSI is used to interface to an IWF this sub-clause does not apply because unit roaming is not in scope of the present Annex.

6.5.3 ISSI Support for Roaming of the Called SU

When the ISSI is used to interface to an IWF this entire sub-clause does not apply because unit roaming is not in scope of the present Annex.

6.5.4 ISSI Support for Roaming of the Calling SU

When the ISSI is used to interface to an IWF this entire sub-clause does not apply because unit roaming is not in scope of the present Annex.

6.5.5 ISSI SU-to-SU Voice Call Release by Calling Serving RFSS

When the ISSI is used to interface to an IWF this entire sub-clause does not apply because unit roaming is not in scope of the present Annex.

6.5.6 ISSI SU-to-SU Voice Call Release by Calling Home RFSS

Items 1, 3, 4 and 5 apply when an ISSI is used to interface to an IWF.

6.5.6.1 Actions Required at Calling Home RFSS

The statement about sending the SIP BYE to the Called Home applies when an ISSI is used to interface to an IWF. The statement about sending the SIP BYE to the Calling Serving does not apply when using the ISSI to interface to an IWF because there is no Calling Serving connected to the IWF via ISSI.

6.5.6.2 Actions Required at Called Home RFSS

This clause applies as written when the ISSI is used to interface to an IWF.

6.5.6.3 Actions Required at Called Serving RFSS

This clause applies as written when the ISSI is used to interface to the IWF.

6.5.6.4 Actions Required at Calling Serving RFSS

This clause does not apply when the ISSI is used to interface to an IWF.

6.5.7 ISSI SU-to-SU Voice Call Release by Called Home RFSS

Item 1 in this section applies when the ISSI is used to interface to an IWF.

For item 2, which refers back to 6.5.6 with “calling” and “called” transposed, items 1, 3, 4 and 5 apply when an ISSI is used to interface to an IWF.

When this clause refers back to 6.5.6 with “calling” and “called” transposed, then the same caveats listed for 6.5.6.1 through 6.5.6.4 above apply, except the statement in 6.5.6.1 about sending the SIP BYE to the Called Serving (“calling” replaced with “called”) does not apply because when using the ISSI to interface to an IWF there is no Called Serving connected to the IWF via ISSI.

6.5.8 ISSI SU-to-SU Voice Call Release by Called Serving RFSS

When the ISSI is used to interface to an IWF this entire sub-clause does not apply because unit roaming is not in scope of the present Annex.

6.6 Call Control Procedures for ISSI Group Voice Services

This entire clause including [102BACA] Addendum 1 sub-clause 6.6.8 ISSI Group Emergency Cancellation is applicable when the ISSI is being used to interface with an IWF.

7. Push-to-Talk (PTT) Management

This entire clause is applicable when the ISSI is being used to interface with an IWF. Note that when the transmission vocoder mode is discussed, which vocoder modes the IWF supports will reflect its capabilities and the capabilities of the MC system.

Limited Use Copy

8. RFSS Capability Polling

This clause is applicable when the ISSI is being used to interface with an IWF

Limited Use Copy

Annex A (Normative) Generic SIP Responses over the ISSI

This entire Annex is applicable when the ISSI is being used to interface with an IWF.

Limited Use Copy

Annex B (Normative) Table of Timers and Constants.

This entire Annex is applicable when the ISSI is being used to interface with an IWF.

Limited Use Copy

Annex C (Informative) – TIA-102 Vocoder Error Mitigation Data

This entire Annex is applicable when the ISSI is being used to interface with an IWF.

Limited Use Copy

Annex D (Informative) – SU-to-SU Call Flows

This Annex's depiction of the interface between the Calling Home and Called Home RFSSs is applicable when the ISSI is being used to interface with an IWF. The other interfaces shown are not applicable because unit roaming is not in scope of this Annex. Also note that example D.10 Roaming is not applicable for the same reason.

Limited Use Copy

Annex E (Informative) Group Call Flows

This annex is applicable when the ISSI is being used to interface with an IWF.

Limited Use Copy

Annex F (Informative) – SU-to-SU Call Flows Involving Multiple Vocoder Modes

This Annex's depiction of the interface between the Calling Home and Called Home RFSSs is applicable when the ISSI is being used to interface with an IWF. The other interfaces shown are not applicable because unit roaming is not in scope of this Annex.

Note that which vocoder modes the IWF supports will reflect its capabilities and the capabilities of the MC system

Limited Use Copy

Annex G (Informative) Group Call Flows Involving Multiple Vocoder Modes

This annex is applicable when the ISSI is being used to interface with an IWF.

Note that which vocoder modes the IWF supports will reflect its capabilities and the capabilities of the MC system.

Limited Use Copy

Annex H (Informative) Future Considerations

This annex is applicable when the ISSI is being used to interface with an IWF.

Limited Use Copy

Annex I (Informative) Console functionality

This annex is applicable when the ISSI is being used to interface with an IWF.

Limited Use Copy

Limited Use Copy

Limited Use Copy

THE TELECOMMUNICATIONS INDUSTRY ASSOCIATION

TIA represents the global information and communications technology (ICT) industry through standards development, advocacy, trade shows, business opportunities, market intelligence and world-wide environmental regulatory analysis. Since 1924, TIA has been enhancing the business environment for broadband, wireless, information technology, cable, satellite, and unified communications.

TIA members' products and services empower communications in every industry and market, including healthcare, education, security, public safety, transportation, government, the utilities. TIA is accredited by the American National Standards Institute (ANSI).



TELECOMMUNICATIONS
INDUSTRY ASSOCIATION

1310 N. COURTHOUSE ROAD, SUITE 890
ARLINGTON VA, 22201; USA

+1.703.907.7700 MAIN
+1.703.907.7727 FAX